

Havyn Nguyen

+1 (714) 208-5805 | havynnguyen9@gmail.com | linkedin.com/in/havynnguyen | github.com/hhavynn | hhavynn.github.io

EDUCATION

University of California - San Diego, B.S. in Mathematics-Computer Science

Sep 2023 - June 2027

Relevant Coursework: Software Engineering, Advanced Data Structures, Design & Analysis of Algorithms, Database Systems

WORK EXPERIENCES

Kyocera International, Inc.

San Diego, CA

Incoming Engineering Intern | Software Engineering, Manufacturing Digitalization

Jun 2026 - Sep 2026

- Developing Python/SQL workflows to clean, transform, and analyze manufacturing datasets for production reporting, traceability, and operational decision-making.
- Building internal reporting tools and dashboards that surfaced production status, process bottlenecks, and manufacturing data trends for engineering stakeholders.

AutomaticGPT.ai

San Diego, CA

Software Engineering Intern | Node.js, Express, React, Supabase, Anthropic/OpenAI API

Apr 2026 - Jun 2026

- Designed and shipped an in-memory agent registry that merged static bot definitions with Supabase metadata across ~153 production bots, enabling live CRUD-triggered bot updates without server restarts.
- Built a provider-routing layer that dispatched Claude requests through the Anthropic SDK while preserving the existing OpenAI execution path, unblocking Claude-configured bot execution with minimal service-layer changes.
- Developed a 3-endpoint Bot Builder REST API for generating, saving, and approving bots using Claude structured tool calls, validated JSON output, Supabase persistence, and Auth0 JWT authorization.
- Fixed an approval-gate bug that allowed unpublished bots to enter the live registry, adding publish-state filtering with backwards-compatible handling for legacy records.

UCSD Chiba Lab

La Jolla, CA

Undergraduate Researcher | Python, pandas, data pipelines, EDA

May 2025 - Feb 2026

- Built Python/pandas pipelines to clean and transform auditory and biometric time-series datasets, improving data readiness for statistical analysis and research EDA.

PROJECTS

VSA at UCSD Website — Full-Stack Web Platform | React, TypeScript, PostgreSQL, Supabase

May 2025

- Built a production React/TypeScript + Supabase platform for 600+ VSA members, supporting public member-facing pages and admin-only workflows for events, attendance, points, and content management.
- Architected PostgreSQL data models for members, event attendance, academic terms, yearly leaderboards, and house standings using RLS policies, constraints, aggregate views, and triggers for secure, consistent data access.
- Developed duplicate-safe Google Sheets/CSV attendance imports with school-email normalization, exact/fuzzy-name matching, manual review, smart merge flows, and import audit logs.

Triton Fetch — Chrome Web Extension | JavaScript, Python, Chrome APIs, yt-dlp

Apr 2026

- Engineered a Chrome MV3 extension that captures authorized UCSD lecture HLS/transcript resources, extracts course and lecture metadata from Canvas/Kaltura pages, and prevents stale async writes using per-tab navigation versioning.
- Built a Python native messaging host implementing Chrome's length-prefixed JSON protocol to orchestrate yt-dlp subprocess downloads, stream real-time progress, and handle ~230MB average lecture video files.
- Added duplicate-download prevention, 2-hour subprocess timeouts, tail-buffered error classification, cross-platform host installers, and sanitized filename generation; validated on 50+ lectures across ~36 UCSD courses.

Matcha Scout — AI Matcha Recommender | Next.js, FastAPI, AWS Lambda, DynamoDB, Docker, Kubernetes

Jun 2026

- Built and deployed an AI-powered matcha recommendation app that converts drink reviews into structured taste profiles and ranks drinks by user preferences across strength, sweetness, creaminess, earthiness, and bitterness.
- Deployed a serverless FastAPI backend on AWS Lambda/API Gateway with DynamoDB persistence, Vercel-hosted Next.js frontend, and production CORS/env configuration for full-stack public access.
- Containerized local development with Docker Compose and authored Kubernetes manifests validated on a local kind cluster, including API/frontend services, DynamoDB Local, seeding, and port-forward smoke tests.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C++, HTML/CSS, JavaScript, TypeScript,

Frameworks & Tools: Node.js, Express.js, REST APIs, PostgreSQL, MySQL, SQLite, Redis, Supabase, JWT, OpenAPI, OpenTelemetry, React, Next.js, Vite, Tailwind CSS, React Query, React Hook, Plotly, Docker, CI/CD, AWS: EC2, S3, Vercel

LEADERSHIP

Triton Web Developers

La Jolla, CA

Developer

Apr 2026 - Present

- Selected as one of 12 members from a competitive applicant pool to join a project-based organization focused on delivering high-quality web solutions for campus clients.

Vietnamese Student Association at UCSD

La Jolla, CA

Co-President

May 2026 - Present

- Lead executive board operations for a 600+ member cultural organization, building on prior Co-Events Chair experience coordinating major events, membership systems, and digital tools across cabinet teams.

Association for Computing Machinery AI

La Jolla, CA

Social Lead

Sep 2025 - May 2026

- Host and coordinate AI-focused community events drawing 100+ students per quarter